DAMIEN CHEMISTRY OLYMPIAD NOTES

Gaseous State

Questions Only

Contents: kinetic theory, ideal gases, partial pressure, gas stoichiometry, non-ideal behaviour and Olympiad extensions

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Data and Constants

Unless otherwise stated, assume that gases behave ideally and use

$$R = 8.31 \ \text{kPa} \ \text{dm}^3\text{mol}^{-1} \ \text{K}^{-1}.$$

Temperature must be expressed on the absolute scale:

$$T(\text{K}) = \theta(^\circ\text{C}) + 273.$$

Conceptual Questions

Question 1: Assumptions of the ideal-gas model

Consider the following statements about an ideal gas.

1. Gas particles have negligible volume compared with the volume of the container.
2. Gas particles exert no intermolecular forces on one another.
3. Gas particles lose kinetic energy whenever they collide.
4. Gas pressure arises from collisions of particles with the walls of the container.
5. Gas particles remain stationary between collisions.

Identify the correct statements and explain why each incorrect statement is incorrect.

Question 2: Temperature and molecular speed

Separate samples of neon and argon are maintained at the same temperature. Compare:

1. their average molecular kinetic energies;

Question 3: Why absolute temperature is required

A student substitutes 25 into the ideal-gas equation for a gas at 25 °C. Explain why this is incorrect.

Question 4: Conditions favouring ideal behaviour

Explain why gases behave most ideally at high temperature and low pressure.

Question 5: Comparing deviations from ideality

Ammonia, methane and helium are examined at the same moderately low temperature and pressure. Which gas would be expected to show the greatest deviation caused by intermolecular attraction? Explain your answer.

Question 6: Compressibility factor

The compressibility factor Z is defined as

$$Z = \frac{pV}{nRT}.$$

Explain what is indicated by each of the following:

1. $Z = 1$;
2. $Z < 1$;
3. $Z > 1$.
4. Interesting Extension Question: The Joule-Thomson Effect is the temperature change when a real gas expands from a high pressure area to a low pressure area without any heat exchange. As the molecules spread apart to fill up the larger area, the potential energy increases. By conservation of eneergy, the kinetic energy decreases and temperature falls. In fact, this is the working principle of refrigerators! In order for this principle to work, what value of Z must the real gas have?

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Question 7: Adding an inert gas

A rigid container contains nitrogen at constant temperature. Helium is added without reacting with the nitrogen. Explain what happens to:
1. The partial pressure of nitrogen;
2. The total pressure.

How would the answer differ if the container could expand while the total pressure remained constant?

Question 8: Gas collected over water

A gas is collected by displacement of water. Explain why the measured pressure cannot normally be used directly as the pressure of the collected gas.

Question 9: Comparing molecular models

Two simplified models are proposed for real gases.
- **Model I:** particles have negligible volume but attract one another.
- **Model II:** particles have finite volume but do not attract one another.

Experimental measurements show that a gas has $Z < 1$ at moderate pressure but $Z > 1$ at very high pressure. Evaluate the two models.

Calculation Questions

Question 10: Determining molar mass

A 0.724 g sample of a volatile substance forms 600 cm^3 of vapour at 101 kPa and 300 K. Determine the molar mass of the substance.

Question 11: Gas density

Calculate the density of sulfur dioxide at 150 kPa and 350 K. Take

$$M(SO_2) = 64.1 \text{ g/mol}.$$

Question 12: Change of gas state

A gas occupies 2.50 dm^3 at 100 kPa and 298 K. Calculate its new volume when the pressure is increased to 160 kPa and the temperature is increased to 350 K.

Question 13: Gas stoichiometry

Magnesium reacts with excess hydrochloric acid according to

$$Mg(s) + 2\ HCl(aq) \rightarrow MgCl_2(aq) + H_2(g).$$

Calculate the volume of hydrogen produced when 0.500 g of magnesium reacts completely at 101 kPa and 298 K. Take $A_{r(Mg)} = 24.3$.

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Question 14: Partial pressures from masses

A 10.0 dm^3 container holds 4.00 g of helium and 10.0 g of neon at 300 K. Calculate:
1. the partial pressure of helium;
2. the partial pressure of neon;
3. the total pressure.

Take $M(\mathrm{He}) = 4.00$ g/mol and $M(\mathrm{Ne}) = 20.2$ g/mol.

Question 15: Gas collected over water

A 250 cm^3 sample of gas is collected over water at 298 K. The total pressure is 102.0 kPa, and the vapour pressure of water at this temperature is 3.17 kPa. Calculate the amount of dry gas collected.

Question 16: Selective absorption

A 5.00 dm^3 container holds 0.0300 mol of $\mathrm{CO_2}$, 0.0200 mol of $\mathrm{O_2}$, and 0.0500 mol of $\mathrm{N_2}$ at 300 K. All the carbon dioxide is selectively absorbed without changing the temperature or container volume.

Calculate:
1. the final total pressure;
2. the final partial pressure of oxygen.

Question 17: Adding another gas

A rigid 10.0 dm^3 container initially contains 0.200 mol of nitrogen and 0.100 mol of oxygen at 300 K. A further 0.100 mol of helium is added without changing the temperature.

Calculate:
1. the initial partial pressures of nitrogen and oxygen;
2. the initial total pressure;
3. the partial pressure of each gas after helium is added;
4. the final total pressure.

Question 18: Identifying an unknown gas

A 0.552 g sample of an unknown gas occupies 250 cm^3 at 100 kPa and 300 K. The gas is one of the following:

$$\mathrm{N_2O}, \quad \mathrm{SO_2}, \quad \mathrm{C_4H_8}, \quad \mathrm{Cl_2}.$$

Identify the gas.

Question 19: Reacting and non-reacting gases

A rigid container contains methane, oxygen and nitrogen in the mole ratio

$$\mathrm{CH_4} : \mathrm{O_2} : \mathrm{N_2} = 1.0 : 2.5 : 1.5.$$

The initial total pressure is 105 kPa. The methane is ignited according to

$$\mathrm{CH_4}(g) + 2\mathrm{O_2}(g) \rightarrow \mathrm{CO_2}(g) + 2\mathrm{H_2O}(g).$$

The mixture is then cooled to its original temperature so that all the water formed condenses. Assuming the volume remains constant, calculate:
1. the final total pressure;
2. the final partial pressures of carbon dioxide, oxygen and nitrogen.

Question 20: Olympiad extension: testing the ideal-gas model

One mole of a real gas is studied at 300 K.
Under condition A:

$$p = 450 \ \text{kPa}, \quad V = 5.00 \ \text{dm}^3.$$

Under condition B:

$$p = 1,000 \ \text{kPa}, \quad V = 3.00 \ \text{dm}^3.$$

Calculate the compressibility factor

$$Z = \frac{pV}{nRT}$$

under each condition. Hence, identify the dominant cause of non-ideal behaviour under each condition.

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